

Homework 7

Name: Diya Jain

1. 1. Yes, it is a field.
2. No, it is not a field.
3. No, it is not a field.

2. 1. $\text{Span}_{\mathbb{R}}((1, 2, 0, 3), (1, 0, 0, 1), (0, 0, 0, 200), (2000, 123123, 0, 0))$

$$x(1, 0, 0, 1) + y(0, 0, 0, 200) + z(2000, 123123, 0, 0) = 0$$

$$(x + 2000z, 123123z, 0, x + 200y) = (0, 0, 0, 0)$$

$$x + 2000z = 0 \rightarrow x = 0$$

$$123123z = 0 \rightarrow z = 0$$

$$x + 200y = 0 \rightarrow y = 0$$

$(1, 2, 0, 3)$ is in the span of $\{(1, 0, 0, 1), (0, 0, 0, 200), (2000, 123123, 0, 0)\}$

$\therefore$ The dimension is therefore 3.

2. $\mathbb{R}^5$ is spanned by the linearly independent vectors $(1, 0, 0, 0, 0)$, $(0, 1, 0, 0, 0)$, $(0, 0, 1, 0, 0)$, $(0, 0, 0, 1, 0)$ and $(0, 0, 0, 0, 1)$, hence the dimension is 5.

3. $\mathbb{R}^{5 \times 5}$ is generated by the 25 linearly independent matrices M_{ij} that have a 1 in the coefficient i, j and zeroes elsewhere.

3. 1. $\{1, 4, 3, 2, 5\} \rightarrow \{1, 2, 3, 4, 5\}$

We obtain reordering by swapping 4 and 2, so the signature is -1.

2. $\{1, 5, 2, 3, 4\} \rightarrow \{1, 2, 5, 3, 4\} \rightarrow \{1, 2, 3, 5, 4\} \rightarrow \{1, 2, 3, 4, 5\}$

We obtain reordering by swapping 2 and 5, then 5 and 3, then 4 and 5, so the signature is -1.

$$3. \{5, 1, 2, 3, 4\} \rightarrow \{1, 5, 2, 3, 4\} \rightarrow \{1, 2, 5, 3, 4\} \rightarrow \{1, 2, 3, 5, 4\} \rightarrow \{1, 2, 3, 4, 5\}$$

We obtain the reordering by swapping 1 and 5, then 5 and 2 then, 5 and 3 then, 4 and 5, so the signature is 1.

$$4. A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$1. \det(A) = (1) \times (-1)^{1+1} \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} + (2) \times (-1)^{1+2} \begin{vmatrix} 0 & 6 \\ 7 & 9 \end{vmatrix} + (0) \times (-1)^{1+3} \begin{vmatrix} 0 & 5 \\ 7 & 8 \end{vmatrix}$$

$$= (45 - 48) + (2)(-1)(0 - 42) + 0$$

$$= (-3) + 84$$

$$= 81$$

$$2. C_{11} = (-1)^{1+1} \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} = (45 - 48) = -3$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 0 & 6 \\ 7 & 9 \end{vmatrix} = (-1)(0 - 42) = 42$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 0 & 5 \\ 7 & 8 \end{vmatrix} = (1)(0 - 35) = -35$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 2 & 0 \\ 8 & 9 \end{vmatrix} = (-1)(18 - 0) = -18$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 0 \\ 7 & 9 \end{vmatrix} = (1)(9 - 0) = 9$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = (-1)(8-14) = 6$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 2 & 0 \\ 5 & 6 \end{vmatrix} = (1)(12-0) = 12$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 0 \\ 0 & 6 \end{vmatrix} = (-1)(6-0) = -6$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ 0 & 5 \end{vmatrix} = (1)(5-0) = 5$$

$$\text{cofactor matrix} = \begin{bmatrix} -3 & 42 & -35 \\ -18 & 9 & 6 \\ 12 & -6 & 5 \end{bmatrix}$$

$$3. \quad A^{-1} = \frac{1}{\det(A)} \cdot C^T = \frac{1}{81} \begin{bmatrix} -3 & -18 & 12 \\ 42 & 9 & -6 \\ -35 & 6 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} -1/27 & -2/9 & 4/27 \\ 14/27 & 1/9 & -2/27 \\ -35/81 & 2/27 & 5/81 \end{bmatrix}$$

$$5. \quad 1. \quad \text{Since } B_1 \text{ is in canonical basis, } x = w_1 + w_3 = \begin{pmatrix} 1 \\ 0 \\ 7 \end{pmatrix} + \begin{pmatrix} 0 \\ 6 \\ 9 \end{pmatrix} = \begin{pmatrix} 1 \\ 6 \\ 16 \end{pmatrix}$$

$$= v_1 + 6v_2 + 16v_3.$$

$$2. A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$y = v_1 + v_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 0 \\ 7 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 6 \\ 9 \end{pmatrix}$$

$$1 = c_1 + 2c_2 \quad \text{--- (1)} \quad \rightarrow \quad 1 - 2c_2 = c_1$$

$$0 = 5c_2 + 6c_3 \quad \text{--- (2)} \quad \rightarrow \quad -\frac{5}{6}c_2 = c_3$$

$$1 = 7c_1 + 8c_2 + 9c_3 \quad \text{--- (3)}$$

Putting c_1 and c_3 in (3)

$$1 = 7(1 - 2c_2) + 8c_2 + 9\left(-\frac{5}{6}c_2\right)$$

$$1 = 7 - 14c_2 + 8c_2 - \frac{15c_2}{2}$$

$$-6 = \frac{-28c_2 + 16c_2 - 15c_2}{2}$$

$$-6 = \frac{-27c_2}{2}$$

$$c_2 = \frac{-12}{-27} = \frac{4}{9}$$

$$1 - 2\left(\frac{4}{9}\right) = c_1$$

$$\frac{1}{9} = c_1$$

$$\frac{-5 \cdot 4}{6 \cdot 9} = \frac{-20}{54} = \frac{10}{27}$$

$$y = \frac{1}{9}w_1 + \frac{4}{9}w_2 + \frac{10}{27}w_3 \quad \text{ans.}$$